

Integrales impropias. Sea $f \in \mathcal{R}$ en $[a, t]$ $\forall t > a$.

$$\int_a^\infty f(x) dx \stackrel{\text{def}}{=} \lim_{t \rightarrow \infty} \int_a^t f(x) dx.$$

Si $f(x)$ tiene antiderivada $F(x)$ en $[a, \infty)$

$$\int_a^\infty f(x) dx = \lim_{t \rightarrow \infty} F(t) - F(a).$$

Problema: $\int_0^x \frac{dt}{2 + \cos^2 t}.$

Sln: $\frac{dt}{2 + \cos^2 t} = \frac{dt}{3 \cos^2 t + 2 \sin^2 t} = \frac{dt}{\cos^2 t} \cdot \frac{1}{(3 + 2 \tan^2 t)}$

$$d(\tan t) = \frac{dt}{\cos^2 t} \quad \text{sea } u = \tan t \quad v = \sqrt{\frac{2}{3}} u.$$

$$F(x) = \int_0^x \frac{dt}{2 + \cos^2 t} = \frac{1}{\sqrt{6}} \int_0^{\sqrt{\frac{2}{3}} \tan x} \frac{dv}{1 + v^2} = \frac{1}{\sqrt{6}} \arctan\left(\sqrt{\frac{2}{3}} \tan x\right).$$

$$f(t) = \frac{1}{2 + \cos^2 t} \quad f(t) \text{ es acotada.} \quad |f(t)| < \frac{1}{3}.$$

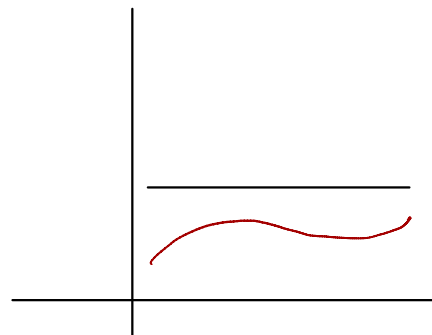
$$|F(x)| < \frac{1}{\sqrt{6}} \frac{x}{2}, \text{ i.e., } F(x) \geq \frac{1}{3} x. \text{ luego.}$$

$A > a$.

$$\int_a^\infty f(x) dx \text{ converge} \Leftrightarrow \int_A^\infty f(x) dx \text{ converge}$$

$$\int_a^\infty f(x) dx \text{ converge} \Leftrightarrow \int_a^\infty c f(x) dx \text{ converge}$$

$$\int_a^\infty f(x) dx, \int_a^\infty g(x) dx < \infty \Rightarrow \int_a^\infty (f(x) \pm g(x)) dx < \infty.$$



Sea $f(x) \geq 0$

$\varphi(t) = \int_a^t f(x) dx$ es creciente.

$\exists \lim_{t \rightarrow \infty} \varphi(t) \Leftrightarrow \varphi(t)$ es acotada.

$f(x) \geq 0$ y $g(x) \geq 0$.

$$i). \left. \begin{array}{l} \int_a^\infty f(t) dt < \infty \\ g(t) \leq f(t) \end{array} \right\} \Rightarrow \int_a^\infty g(t) dt < \infty.$$

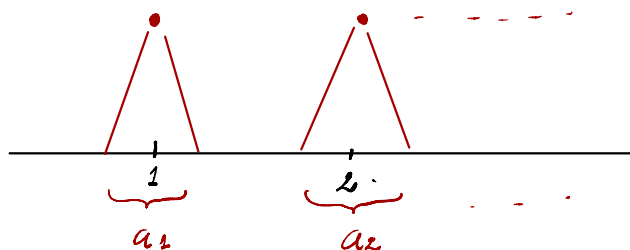
ii). $\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = k \neq 0$ entonces $\int_a^\infty f(t) dt, \int_a^\infty g(t) dt$ convergen simultáneamente.

Problema 1. Sea f continua en $[0, \infty)$.

$\int_0^\infty f(x) dx$ converge $\Rightarrow \lim_{x \rightarrow \infty} f(x) = 0$. {Verdad o Falso}.

Falso: Sea $\sum_{n=1}^\infty a_n$ una serie convergente de términos positivos

con triángulos de base a_n y las alturas igual a $\frac{1}{2}$.



Sea A_n el área del triángulo con base A_n .

$$\int_0^\infty f(x) dx = \sum_{n=1}^\infty A_n = \sum_{n=1}^\infty \frac{a_n}{4} < \infty. \text{ Sin embargo } \lim_{x \rightarrow \infty} f(x) \neq 0.$$

Problema 2. Presentar ejemplo de $f(x)$ continua, no acotada t.q $\int_0^\infty f(x) dx$ converge. y otro que converge $\int_0^\infty f(x) dx = 0$.

Integrales de función no acotada

$f(x)$ integrable en $[a, t]$ $\forall t < b$

Thm. $\int_a^\infty f(x) dx < \infty \Leftrightarrow \forall \varepsilon > 0 \exists \delta > 0$ t.q $A_1 > A_2 > \delta \Rightarrow \left| \int_{A_1}^{A_2} f(x) dx \right| < \varepsilon$.

Thm. $\int_a^\infty |f(x)| dx$ converge entonces $\int_a^\infty f(x) dx$ converge.

Proof. (Para usted).

Thm (Valor medio) $f(x) \geq 0$ decreciente en $[a, b]$ $g(x)$ es integrable en $[a, b]$

entonces $\exists \xi \in [a, b]$ t.q

$$I = \int_a^b f(x)g(x)dx = f(a) \int_a^b g(x)dx.$$

Proof. Sea $\{x_0, \dots, x_n\}$ -particion de $[a, b]$.

$$I = \sum_{i=0}^{n-1} \int_{x_i}^{x_{i+1}} f(x)g(x)dx = \sigma + \rho.$$

$$\sigma = \sum_{i=0}^{n-1} f(x_i) \int_{x_i}^{x_{i+1}} g(x)dx$$

$$\rho = \sum_{i=0}^{n-1} \int_{x_i}^{x_{i+1}} (f(x) - f(x_i))g(x)dx$$

$|g(x)| \leq L$ en $[a, b]$.

$$|f(x) - f(x_i)| \leq M_i - m_i$$

$$M_i = \sup_{x \in \Delta x_i} f(x) \quad m_i = \inf_{x \in \Delta x_i} f(x)$$

$$|\rho| \leq L \sum_{i=0}^{n-1} (M_i - m_i) \Delta x_i \Rightarrow \lim_{|\rho| \rightarrow 0} \rho = 0.$$

$G(x) = \int_a^x g(t)dt$ - continua en $[a, b]$.

$$\sigma = \sum_{i=0}^{n-1} f(x_i) (G(x_{i+1}) - G(x_i))$$

$$= \sum_{i=0}^{n-1} f(x_i) G(x_{i+1}) - \sum_{i=0}^{n-1} f(x_i) G(x_i)$$

$$= \sum_{i=1}^{n-1} \underbrace{(f(x_{i-1}) - f(x_i))}_{\geq 0} G(x_i) + \underbrace{f(x_{n-1})}_{\geq 0} G(x_n)$$

$$M = \sup_{x \in [a, b]} G(x)$$

$$m = \inf_{x \in [a, b]} G(x)$$

$$\sigma \leq M \left(\sum_{i=1}^{n-1} (f(x_{i-1}) - f(x_i)) + f(x_{n-1}) \right) = M f(a).$$

$$0 \leq m f(a)$$

$$\text{luego } m f(a) \leq I \leq M f(a)$$

$$\exists \mu \in [m, M] \text{ t.q. } I = \mu f(a)$$

$$\exists \xi \in [a, b] \text{ t.q. } \mu = G(\xi).$$

Thm. Sea f monotonu en $[a, \infty)$ y acotada. Si $\int_a^\infty g(x) dx < \infty$ entonces $\int_a^\infty f(x)g(x) dx < \infty$.

Proof. (Criterio Abel).

$$\left| \int_{A_1}^{A_2} f(x)g(x) dx \right| = f(A_1) \int_{A_1}^\xi g(x) dx + f(A_2) \int_\xi^{A_2} g(x) dx \\ \leq 2L\varepsilon \text{ donde } |f(x)| < L.$$

$$\exists \delta \text{ t.q. } a_2 > a_1 > \delta \text{ implica } \left| \int_{a_1}^{a_2} g(x) dx \right| < \varepsilon. \quad \square$$

Hemos usado que:

Si f es creciente y no negativa en $[a, b]$ y $g(x)$ es integrable en $[a, b] \Rightarrow \exists \xi \in [a, b] \text{ t.q.}$

$$I = \int_a^b f(x)g(x) dx = f(b) \int_\xi^b g(x) dx.$$

$$\text{y adem\'as } \int_a^b f(x)g(x) dx = f(a) \int_a^\xi g(x) dx + f(b) \int_\xi^b g(x) dx.$$

Criterio Dirichlet. Sea f monotonu, $\lim_{x \rightarrow \infty} f(x) = 0$, $g(x)$ integrable en $[a, b]$ y $\left| \int_a^b g(x) dx \right| < K \quad \forall b$. Entonces $\int_a^\infty f(x)g(x) dx < \infty$.

Proof. (Para usted).

Def: $\int_{-\infty}^{\infty} f(x) dx \stackrel{\text{def}}{=} \lim_{\substack{A \rightarrow +\infty \\ B \rightarrow -\infty}} \int_B^A f(x) dx = \lim_{A \rightarrow \infty} \int_a^A f(x) dx + \int_B^a f(x) dx.$

$\neq \lim_{A \rightarrow +\infty} \int_{-A}^A f(x) dx \stackrel{\text{def}}{=} \text{v.p} \int_{-\infty}^{+\infty} f(x) dx.$

$\forall f$ impar $\text{v.p} \int_{-\infty}^{+\infty} f(x) dx = 0$ sin embargo $\nexists \int_{-\infty}^{\infty} x dx.$

Problema: Si $\int_{-\infty}^{+\infty} f(x) dx = A < \infty$ entonces $\text{v.p} \int_{-\infty}^{+\infty} f(x) dx = A.$